Bethlendi *et al.*
*Journal of Innovation and Entrepreneurship* (2025) 14:35
https://doi.org/10.1186/s13731-025-00493-w

Journal of Innovation and Entrepreneurship

RESEARCH Open Access

# What could we learn from startup failures?

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**Abstract**

This study aimed to empirically investigate resource-based and dynamic capability theories in startup failures, categorize factors that contributed to failures, and study the gender effect on startup failure. Data consists of 40 information and communication technology startups in the voluntary liquidation phase. The questionnaire technique was developed based on SHELL model. Research questions were investigated using principal component analysis, Ward's hierarchical cluster, and cross-table analysis. The main identified groups of startup failure factors are mostly dynamic capabilities, supporting dynamic capability over a resource-based view. The lack of dynamic capabilities is clearly identifiable in one subgroup of failed startups. The relevance of some failure factors changes with the life cycle of startups. This study found a positive relationship between the presence of female founders and the startup survival. Besides, it highlights the importance of innovation and corporate foresight training programs, mentorship access, and the need to strengthen project screening standards by focusing more on dynamic capabilities. This study underpins the necessity to develop policies easing the gender gap. These would improve the efficiency of the startup ecosystem and entail positive social effects.

**Keywords:** Startup success, Startup failure, Dynamic capabilities theory, Resource-based view, SHELL methodology, Gender

## Introduction

Startups typically aspire to rapid, international growth and develop pioneering, innovative products. This behavior carries a significant business risk, resulting in a high failure rate. However, successful ones can bring high returns to investors, but this operating model also requires specific funding resources. Venture capital (VC) is the most important funding type of startups. In Europe, there is a policy goal to close the VC investment gap compared with the US (European Commission, 2021). The availability of these resources has been hectic in recent years. According to KPMG's global survey, venture financing nearly doubled between 2018 and 2021. In 2021, the total global venture capital financing was USD 749 billion. Then it fell back to 2018 levels by 2024 (KPMG, 2024). In many countries, significant public resources have also been channeled in addition to private investments since policymakers expect innovation-driven economic growth from startups. Combined with grants, subsidized loans facilitate turning research results into marketable products through investments in tangible assets. Startups that participate in both types of programs (private and public) outperform grant-only recipients in

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terms of innovation performance, employment, and future revenues (Hottenrott & Richstein, 2020).

Studies also show that many important innovations in recent decades have been introduced not by existing companies but by startups (Angelidou et al., 2018; Anthony et al. 2018). This literature focuses more on successful startups (unicorns, big exits) although this is only a small percentage of cases. By this approach, they omit the predominant part of the available information and experiences. Thus, we are examining failed startups. Our goal is to draw general rules that can be used to improve the very low success rate in the startup ecosystem. Guzman and Stern (2016) demonstrate the importance of improving entrepreneurial quality, resulting in higher startup survival rates, which would have a significant effect on US economic growth.

The reasons for failure can be diverse, and no generally accepted theory exists according to the literature. The resource-based theory (RBV) introduced by Wernerfelt (1984) could explain small businesses' success or failure factors. Since startups often face resource shortages, this theory might help understand how startups' resources drive performance. Due to the limitations of financial data in the case of young firms, non-financial factors have a pivotal role in determining survival (e.g., Lussier, 1995, 1996). According to Teece et al. (1997), RBV does not adequately explain the competitive advantage of a dynamic market. Thus, a new concept, the dynamic capabilities theory, emerged. The literature analysis identified several studies that analyze startups either from a resource-based view (e.gColeman et al., 2013; Zahra, 2021) or from a separate dynamic capability-based view (e.g.: Boccardelli & Magnusson, 2006; Teixeira et al., 2021). There are surprisingly few investigations to link the two views. The most significant is the study of McKelvie and Davidsson (2009). These examined how founders' and firms' resources affect the development of dynamic capabilities in new firms. Different resource types have diverse impacts on different types of dynamic capabilities. For example, the level of education has a positive effect on new product or process development capabilities but does not affect idea generation capability. We aim to empirically investigate if factors of RBV or dynamic describe better new firms (startup) failures.

After Cantamessa et al. (2018), we use the SHELL methodology to plan the semi-structured interview to identify startup failure causes in a structured way. The interview-based approach is in line with the literature since most of the relevant literatures rely on the analysis of interview and questionnaire surveys. These studies cover databases of between 5 and 50 startups and startup founders. Our sample contains 40 information and communications technology (ICT) startups in the voluntary liquidation phase. We interviewed startup founders after their failures and analyze their achievements. Thus, our sample size is in line with the relevant literature. Startups investigated had a Hungarian headquarters. The Central and Eastern European regions are understudied in the empirical literature. The sample size is representative, covering 34% of the total Hungarian startup divestments in the period examined.

Dominant success/failure factors might change by startup life cycle, which is poorly investigated by the literature. Thus, we considered it important to introduce some life cycle variables to capture the various steps of the startup business development.

In entrepreneurship, the concept of entrepreneurial orientation emerged since its role is vital for firms in the startup phase. Entrepreneurial orientation (EO) means the

capability to cover the skills and knowledge gap to boost the startup's success. The link between dynamic capabilities and entrepreneurial orientation has a significant research interest (Coelho et al., 2022; Kee & Rahman, 2018; Monteiro et al., 2017). The gender difference in Entrepreneurial Orientation has a growing interest as well (Kee & Rahman, 2018). Female involvement is not only a social question but might influence economic performance (Ong et al., 2016).

The main purposes of this study are:

1. to identify and group the different factors of failure and failed startup types, empirical investigation of resource-based and dynamic capabilities theories;
2. to investigate whether these failure factors depend on the startup life cycle;
3. to analyze separately the gender effect in business failure;

We proceed as follows: in "Literature review" section, we will review the theoretical background of our study and empirical literature on the factors influencing success and failure of startups and the potential applications of the SHELL model. This is followed in "Data and methodology" section by a presentation of the database of the startups studied and the research methodologies used for their analysis. "Results" section presents the results using descriptive statistics of the variables, principal component analysis, and cluster analysis. We conclude our paper by identifying the discussion points and conclusions, as well as policy implications (Discussion and conclusions and Policy implications sections).

## Literature review

### Defining what is a startup

There is no one commonly accepted definition for startups. Related specifications try to capture some particular nature and characteristics of these businesses. One of the most important values of startup companies is that they are fast-growth-oriented. They achieve the velocity with money provided by forms of financing (business angels, seed capital companies, crowdsourcing platforms, etc.), which are not available for traditional companies (Condom-Vilá, 2020). Another important feature is that these entrepreneurs constantly seek changes or make the most appropriate strategic choices as a means of overcoming problems, leading to business success (Hormiga et al., 2018). One of the most widely adopted definitions of these firms is that a startup is a company designed to achieve a repeatable and scalable business model (Blank, 2010; Clarysse & Bruneel, 2007). Startups that are able to develop solutions that can reach global audiences in a relatively short time and with outstanding profitability are also able to create exceptional business value (Linina et al., 2018). Indeed, this literature suggests that an important feature of startups from the earliest stages is an attitude and a business model that allows for rapid and international growth.

Also, the fundamental nature of startups is innovation to create new products or services, which carries significant risk (Bethlendi, 2019). Schumpeter (1934) was the theoretical precursor who pointed out the relationship between entrepreneurship and innovation, especially as these categories determine growth and economic development. Innovation capability is often addressed implicitly in the literature, and it is seen as a

firm's attitude to turn market opportunities into real advantages by identifying new ideas (Aas & Breunig, 2017) and transforming them into new products, services, or processes with economic value (Noordin & Mohtar, 2013).

The factors affecting the success of startups have been widely studied in the literature over the past few years. Skawinska and Zalewski (2020) found in the European Union that more developed countries offer an institutional competitive advantage to startups, while the gap in success factors between highly developed and catching-up countries can be attributed to human capital and institutions. Argaw and Liu's research indicates that government support plays a greater role in emerging markets, reflecting the need for agility and reliance on local initiatives due to regulatory and financial constraints (Argaw & Liu, 2024). Some of the literatures also examine how the factors affecting the success of startups correlate with their life stage. Barz and co-authors concluded that Critical Success Factors (CFFs) of Networks, Funding, and Responsiveness are important in the startup and scale-up stages. While the factors of Entrepreneurial Culture and Organizational Governance become critical in the mature stage (Barz et al., 2023). According to another approach, critical success factors can be categorized under five broad themes: actor's knowledge and communication process, service value and effectiveness, robust technological infrastructure, revenue generation ability, and regulation management capacity Chakraborty et al., 2023).

Another important attribute is that the majority of startups fail in their very early stages of operation (Griffin, 2017) and various statistics indicate that the failure rates are around 90% (Giardino et al., 2014; Startup Genome 2018, 2022). In this context, we have reviewed the literature on the factors affecting the viability of startups in particular.

#### Conceptual framework for success and failure factors of startups

The resource-based view (RBV) of Wernerfelt (1984) could provide a theoretical background to explain the success or failure factors of small businesses. This theory suggests that a firm's competitive advantage stems from the resources it controls. These resources (physical assets, intellectual property, human and financial resources, and organizational capabilities) are seen as the key to achieving superior performance. The competitive advantage is sustained as long as the firm maintains its control over valuable, rare, and inimitable resources. Newbert (2007) summarized the empirical studies of the resource-based view and found that roughly half of them support the theory empirically, which is the appropriate ratio in management science. However, he pointed out that firms' valuable and unique capabilities and core competencies are essential to get a competitive edge over their static resources. Recently, on the basis of RBV, a new concept, the dynamic capabilities theory, emerged because RBV does not adequately explain the competitive advantage in a dynamic market (Teece et al., 1997). Early proponents of dynamic capabilities argued that the availability of resources alone is not enough for success, transformative capabilities are needed to coordinate and use resources to achieve the firm's goals (Amit & Schoemaker, 1993). The dynamic capability is a startup's competence to create, extend, and manage its resource base (Monferrer et al., 2015). Managing tangible and intangible resources supports startups" survival (Pigola et al., 2022). Dynamic capabilities emphasize processes (how the firm develops, integrates, and reconfigures its resources) rather than the static possession of resources. Dynamic

capabilities emphasize processes, which make adaptable firms allowing them to innovate and adjust strategies to maintain competitive advantage over time. According to Eisenhardt and Martin (2000), dynamic capabilities are specific, well-known business and strategic value-creating processes such as product development or process development capabilities.

Lussier (1995 and 1996) introduced a model primarily relying on non-financial factors to determine the chance of a young firm survival. Due to the limited number of financial data of young firms, non-financial factors have a pivotal role. The Lussier model consists of 15 mostly non-financial variables that were selected from previous studies. Most of them are resource-type variables in line with resource-based theory. Gyimah et al. (2019) summarize the significance of the Lussier model's variables by analyzing 40 articles. They found very mixed results. They were not able to clearly identify a list of variables that contribute with high significance to the success or failure of small businesses. In addition, the success of startups depends more on their dynamic capabilities since all of them face significant resource constraints. Thus, we want to go beyond a narrow resource-based approach and incorporate some strategic business components and dynamic capabilities (e.g., product development, business model fitting, etc.) as well, which are key for the success.

Cantamessa et al. (2018) applied a modified model of SHELL methodology, which is an error-centered approach to the causes of (startup) failures. The original methodology is a well-known approach in the aviation sector for describing accident causes and was developed by Hawkins in 1975 based on the original work proposed by Edwards in 1972 under the name SHEL model. (Hawkins, 1993) The letters in the acronym stand for the following topics: Software, Hardware, Environment, Liveware People, and Liveware Environment. Due to this human-centered structure, the SHELL model is defined as the relationship between human factors and the aviation environment (Reinhart, 1996). The model adopts a system perspective that suggests that the human being is rarely the only cause of an accident. In fact, there are different factors interacting with the human operator and affecting performance. Thus, the SHELL model considers both active failures and latent failures. Active failures typically occur at the operational end, while latent failures are hidden in the organization's structure, not known or observed by the organization (Swedish Club Academy, 2011). Cantamessa et al. (2018) modified this model in their research to fit the problem of entrepreneurial outcomes. In their adaptation, the business model is analogous to the software (S), the product to the hardware (H), the startup's environment to the accident environment (E), the organization to the central liveware (L), and the customer/user to the liveware (L). This led to a repeatable and scalable methodology that can be applied to a database of unstructured post-mortem documents deriving startup failure patterns (Cantamessa et al., 2018).

In the modified SHELL model by Öndas and Akpinar (2021), managerial issues align with the software component, product-related problems with the hardware component, financial challenges with the environmental component, and market-related problems with the liveware component. Cantamessa et al. (2018) conclude that the absence of a structured business development strategy is a pivotal predictor of startup failure, while Öndas and Akpinar provide recommendations to address concerns related to each component of their model (Szathmári et al., 2024).

### Summary of empirical literature

It is widely agreed in the literature that the failure of a startup is hardly ever attributable to one single factor. (Cantamessa et al., 2018). Several research studies summarize that the main preconditions for creating a successful startup are an innovative and competitive idea, a supportive business and political environment, and the idea of skills and competencies to be successfully implemented (van Weele et al., 2018; Zhou & Verburg, 2020). A similar conclusion was reached in another study, which found that four factors have a significant influence on the two ways of measuring success: the location of the startup, the promoting partners' dedication, the age of the company, and the existence of non-promoting partners (Díaz-Santamaría & Bulchand-Gidumal, 2021). According to research by Binowo and Hidayanto (2023), problem and team factors are the most dominant and influential factors that influence the success of a digital startup. Problem factor means whether digital startup founders can detect consumer problems and develop solutions to these problems. Then, the team can quickly and efficiently resolve these problems.

It has also been found that the gender entrepreneurship gap in startups, which represents the difference between the startup founding rates of men and women, is an issue of growing academic and societal interest. Given the growing role of startups in the economic development potential, the question of whether a part of society is excluded from this competition is not negligible. Related research has shown that the gender gap in entrepreneurship persists even within developed economies where gender gaps in wages or professional development have declined (Global Entrepreneurship Monitor, 2017). Previous analyses have also identified a wide range of factors that contribute to the presence and persistence of the gender gap in entrepreneurship, such as differences in educational preferences (Card & Payne, 2021), a lack of female mentors (Rocha & Van Praag, 2020), and unequal opportunities to access to funding sources (Lee & Huang, 2018). The research of Miric et al. (2022) found that serial entrepreneurship (i.e., founding multiple businesses sequentially) is also more common among male founders, which contributes to the gender gap in entrepreneurship.

There is also a growing literature on how industry and regional specificities influence female-founded startups. Kalnins and Williams (2014) highlighted that female-owned businesses out-survive male-owned businesses in a wide variety of industries, most notably in educational services. In terms of geographic area, female-owned businesses consistently out-survived male-owned businesses in the largest cities. Related research also indicates that the most influential parameters are at the local level in the startup ecosystems, rather than at the national level. Local public policies could address the female labor participation rate, especially in startups, and thereby increase the pool of entrepreneurial talent (Berger & Kuckertz, 2016). Given that a significant proportion of startups are founded in the tech sector, it is interesting to know how female founders view these industries. The most common motivators for women to become interested in the tech sector and to become entrepreneurs were related to being able to make something new, innovate, and gain creative freedom to do whatever they wanted. On the other hand, the most common reasons why women chose not to try entrepreneurship were related to the threat of financial risk(s), lack of support from the social environment, and lack of suitable examples and success

stories (Kovaleva et al., 2023). Some researches focus on mechanisms by which female founders might achieve higher performance. Avnimelech and Rechter (2023) demonstrated that female founders seek and gain more entrepreneurial knowledge, network building, and entrepreneurial self-efficacy during their participation in accelerators than male founders. Another important aspect of women entrepreneurs' performance is their ability to see broader, socially important goals as part of their business mission. Related research has shown that working for social good is an encouraging factor in an entrepreneurial career for women. This corresponds to the findings that women have a higher interest in achieving sustainability in their future careers than men (Gender Equality Index, 2020; Kovaleva et al., 2023).

In summary, most of the previous studies on gender analysis from the 1990s to the early 2000s showed that male entrepreneurs had a greater chance of being more successful than female entrepreneurs (Bertoncelj & Kovač, 2009; Kirkwood, 2009). However, specifically from 2010 onwards, in contrast to previous literature on gender differences, studies found that female entrepreneurs tended to have higher performance improvement (Kee & Rahman, 2018; Ong et al., 2016). These authors noted that female entrepreneurs had a higher motivation to survive, were willing to learn about entrepreneurial knowledge, and possessed better financial control. Qualitative research conducted in Japan and China demonstrates that women's motivations, institutional factors, and cultural elements contribute significantly to the success of female-led ventures (Sequeira et al., 2016). It can, therefore, be stated that reducing gender inequality is an important economic policy issue, as the low participation of women in startups is a wasted resource for society. Women entrepreneurs play a crucial role in improving economic welfare and fostering societal development, particularly in countries with strong institutional support (Terjesen & Lloyd, 2015).

We summarized the ten most relevant articles, which examined the success and failure of startups and their different factors and methods (Table 1).

Most of the papers investigated are purely empirical investigations without any theoretical background. In comparison, only 40% of the literature reviewed included any kind of theoretical framework for a deeper understanding of the drivers. The implementation of the resource-based view (RBV) or the dynamic capabilities theory, which provides a basis for the analysis of the competitive advantage in a dynamic market, allowed a deeper analysis. We did not find any article that compared the application of two theories in startup failures or success.

A major part of the relevant literature relies on the analysis of interview and questionnaire survey responses. In these cases, the researchers have typically tried to identify the most important factors by categorizing and prioritizing the responses received from startups. The authors typically used various qualitative comparative analysis, thematic analysis (TA), and classification techniques to identify the different failure factors. For the other 20% of the studies, a quantitative methodology was used to analyze some external datasets to find determinant factors.

Studies that attempt to identify success factors based on interview and questionnaire surveys commonly cover databases of between 5 and 50 startups and startup founders. Larger databases, typically of one to two hundred startups, are applied in a minority of studies using quantitative models. In the CEE region, we found only one relevant study.

**Table 1** Relevant literature on the success and failure of startups. Source: own research, 2024

| Authors | Research goal: success/failure | Used theoretical framework (RBV/dynamic or other) | Database and data source | Methodology | Key findings |
|---|---|---|---|---|---|
| Bednár and Tarišková (2017) | Failure | None | 51 startups | Questionnaire | Identified startup failure factors: incorrect product pricing, poor cost estimates, lack of capital for further development, lack of market need, and team inability to develop the right MVP or business model |
| Binowo and Hidaynato (2023) | Success | None | 10 digital startup founders | Interview | Identified startup success factors: business ideas, teams, business models, capital or funding, products, incubators, validation, competitors, marketing, technology mastery, market analysis, partners, and passion |
| Cantamessa et al. (2018) | Failure | RBV | 214 high-tech startup | Quantitative modeling (SHELL model) | The lack of a structured business development strategy is the key determinant of startup failure |
| Chakraborty et al. (2023) | Success | Dynamic | 17 stakeholders form healthcare startups | Interview | The identified startup success factor is the entrepreneurs' experience in fulfilling essential aspects like clinicians' needs, consumers' requirements, market demands, and policy requisites |
| Díaz-Santamaría and Bulchand-Gidumal (2021) | Success | None | 50 interviews with entrepreneurs and 340 technology startup responses | Interview | Identified startup success factors: the location of the startup, the promoting partners' dedication, the age of the company, and the existence of non-promoting partners |

**Table 1** (continued)

| Authors | Research goal: success/failure | Used theoretical framework (RBV/dynamic or other) | Database and data source | Methodology | Key findings |
|---|---|---|---|---|---|
| Goswami et al. (2023) | Failure | None | 165 startups | Quantitative modeling | The dearth of capital / running out of money, and inadequate sales and marketing strategy, lead businesses to fall behind rivals |
| Öndas and Akpinar (2021) | Failure | RBV | 16 high-tech startups | Interview (SHELL model) | The causes of startup failure can be classified into four main categories: product, market, financial, and managerial problems |
| Pham and Pham (2021) | Success | None | 5 e-commerce startups founders | Interview | Four factors have significant contribution to e-commerce startups' success: risk-taking propensity, achievement orientation, reliability of e-service quality, and e- networking. Among the four, risk-taking propensity is the most contributing factor |
| Prohorovs et al. (2019) | Success & failure | None | 40 startups | Interview | Companies that tend to be more successful in attracting seed funding are managed by entrepreneurs who have had previous experience in creating business entities and are capable of building a team with employees who have appropriate experience, specialized education, and high-level management skills |

**Table 1** (continued)

| Authors | Research goal: success/failure | Used theoretical framework (RBV/dynamic or other) | Database and data source | Methodology | Key findings |
|---|---|---|---|---|---|
| Szathmári et al. (2024) | Failure | None | 50 startup failure cases | Quantitative modeling | Two prominent competency deficits are pivotal determinants of startup failures: information-seeking and customer service orientation. Additionally, deficits in technical expertise, analytical thinking, and flexibility emerged as significant factors contributing to these failures |

**Table 2** External and internal determinants of the success or failure. Source: own research, 2024

| Internal factors | External factors |
|---|---|
| Technological excellence/mastery | Availability of funding sources |
| Market knowledge | Location of the startup |
| Startup management knowledge | Incubators |
| Personality characteristics of the founders | Investors validation and evaluation processes |
| Idea generation capability | Partnerships |
| Product development capability | Investors validation and evaluation processes |
| Sales and marketing performance | Market and competition |
| Business development performance | Regulation, state support |

The factors that influence the failure of startups can be separated into internal factors (e.g., personal characteristics of founders, skill sets, development strategies) and external factors (e.g., characteristics of the ecosystem that supports startups). Table 2 summarizes the main internal or external determinants of the success or failure of startups found in the literature.

Based on the literature review, it can be concluded that in recent years, there have been several analyses of the external and internal factors associated with the failure of startups.

Considering that a minority of the studies reviewed have applied frameworks to assessing the success of startups. We regarded it important to adopt a model (SHELL) to our research to achieve depth results.

## Data and methodology

The description of the sample and the research design used are described below.

### Presentation of the sample

In our research, we had the opportunity to analyze the performance and failure of 40 Hungarian startups. In Table 1, we summarized the ten most relevant articles, which examined the success or failure of startups and their factors, using different methods. Of the 10 studies examined, 8 had a database of 10–100 startups. Four of these had databases between 40 and 51 and for these factors. Thus, we considered our 40-item database to be appropriate. Annual statistics from The Hungarian Venture Capital and Private Equity Association (HVCA) was also reviewed to assess the appropriateness of the size of our research database. In the period 2020–23, there have been 117 startup divestments in Hungary. The 40 failed startups under voluntary liquidation between 2020 and 2023 we surveyed thus represent 34% of the total population, which we considered to be reasonably representative. The size of the dataset was also considered to be sufficient in comparison with other databases in the reviewed literature. A significant proportion of 40 defunct startups received funding from a public VC fund. All companies were in the ICT sector, with a predominant share of SaaS activities (50%), but there was also a high share of HealthTech companies (20%) and EdTech companies (7.5%) in the sample. Hungary is a mid-sized EU member country. We chose the ICT sector because this is a dynamic and one of the most international sectors. Nearly 20% of the startups in the sample targeted the international market right from the start. This might help extrapolate results to a broader context.

During the interview process, 40 ICT startups were included in the research, all of which were in the process of voluntary liquidation. During the voluntary liquidation, assets, including intellectual property (IP), should be valued and removed (in the form of sales or write-offs) from the books. In the case of IP, market value cannot be directly observed, therefore an external valuer is required. The external valuer does the valuation in a structured way. The evaluation addresses the reasons for the failure, as well as its external and internal factors. It was easy to supplement this process with some specific questions. As for the industry, we were looking for one of the most international helping generalize our results. We chose ICT. In addition, this sector attracts a lot of venture capital. Thus, we were able to collect a larger sample.

The structure of the interviews focused on problems related to the business model, financing difficulties, and challenges in management and product development. When the phenomenon was observed, the value of the variable was set to "1", while in its absence it was set to "0". Thus, the SHELL variables were operationalized as dummy variables, which ensured the complete exclusion of data gaps. In addition, the data for the variables measuring financial performance, such as invested capital and cumulative turnover, were normalized to avoid the distorting effect of outliers in the data set.

### Research objectives and methodology

Our research process consisted of three research objectives:

1. To identify and aggregate failure factors and failed startup types; empirical investigation of resource-based and dynamic capabilities theories.

2. To differentiate failure factors and failed startup types by stages of the startup life cycle.
3. To investigate the effect of gender on business failure.

In the voluntary liquidation process, interviews were conducted with the founders to identify the factors leading to failure. Based on the insights provided by the founders, further statistical analyses were performed to determine the generalizable causes of failure.

The research objectives and methodologies used are summarized in Table 3. Our analyses were conducted using the SPSS software package.

In our hierarchical clustering analysis, we employed Ward's method. This method offers several advantages over alternative clustering techniques. For instance, k-means clustering heavily depends on the initial selection of cluster centroids, which can lead to instability in the results. Additionally, k-means assumes that clusters are spherical, which often does not reflect the true structure of the data, and it is primarily suitable for numerical data. In contrast, divisive clustering methods are more computationally intensive and complex, and may yield less accurate results, especially with large datasets as they start from the entire data set.

The primary benefits of Ward's method include minimizing intra-cluster variance and providing a more accurate depiction of the internal structure of each cluster. Furthermore, it facilitates the visual representation of relationships between clusters through dendrograms, which significantly aids in the interpretation of results. Its robustness and reliability in handling a wide range of variables make it an ideal choice for our research.

**Table 3** Research objectives and methodology. Source: own research, 2024

| Research objective | Applied methodology | The objective to be achieved |
|---|---|---|
| 1. The problem map and startup types | Principal component analysis (varimax rotation) is used to narrow down the number of variables using an established statistical method. In doing so, principal component analysis using the varimax procedure was applied<br>We use Ward's hierarchical cluster analysis to identify failed startup types | We are narrowing the scope of variables and identifying problems in a sample<br>Typification of failed startups |
| 2. Failure factors and startup types in terms of life cycles | We use cross-table analysis for the life cycle of variables and startup types | To identify at which stage:<br>• failure factors are the most typical;<br>• the startup cluster ceased to exist, and how far did they develop |
| 3. The role of gender in the survival of startups | Cross-table analysis between gender and life cycle variables<br>Pearson's $chi^2$ test is used to capture the correlation between variables | To identify a correlation between women being part of the founding team and termination at different stages |
| 4. Exit chances of a failed startup | Analysis of variance, cross-table analysis<br>Pearson's $chi^2$ test was used to test for correlation between nominal variables, while the *F*-test was used to test for significant differences between nominal and metric variables | To analyze how companies that received an exit offer were more successful. If the gender influences exit offers as well |

Using this method, we identified three distinct clusters, each representing different patterns of startup failures. This allowed us to gain deeper insights into the underlying causes of failure based on the variables derived from principal component analysis. Overall, Ward's method contributed to the structured and reliable analysis of the data, which was one of the fundamental objectives of our study. It ensured that the internal homogeneity of each cluster remained high while maintaining well-defined differences between clusters.

## Results

We present the descriptive statistics of variables and our main findings through our three research questions.

### Descriptive statistics of variables

As part of our analysis, we have added a few points to the SHELL model of Cantamessa et al. (2018). (Fig. 1.) In the business model (S) block, we added the variables "Wrong customer development" and "Has the business model changed?". The variables of "COVID, war, macroeconomic problems", "Slow decision process of VC", "Other" and "Legislation" were added to the Environment (E) block. (Fig. 2)

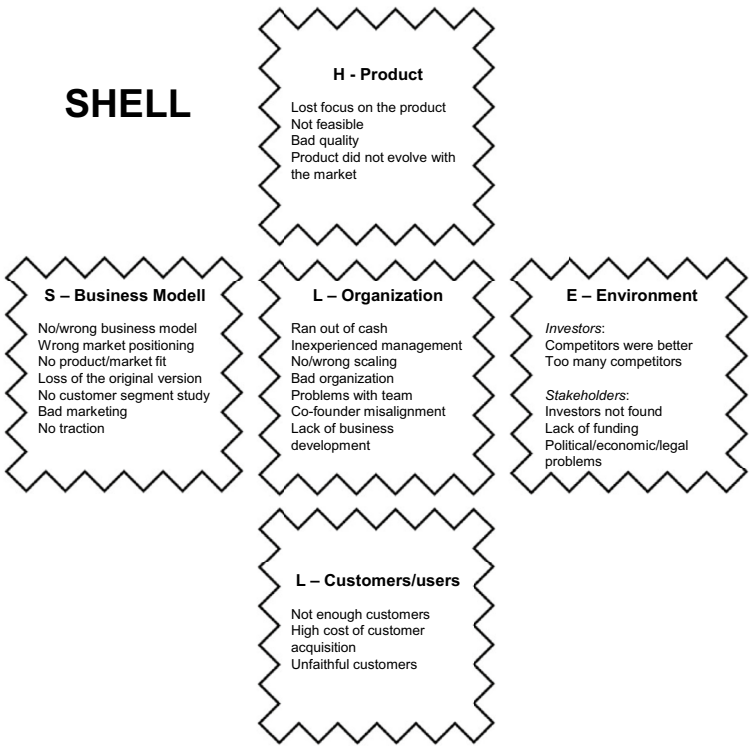



**Fig. 1** A modified version of the SHELL model applied to startups. This simple block list diagram includes the five elements of the SHELL model (Software, Hardware, Environment, Liveware People, and Liveware Environment) with startup-specific problem factors (Source: Cantamessa et al., 2018)

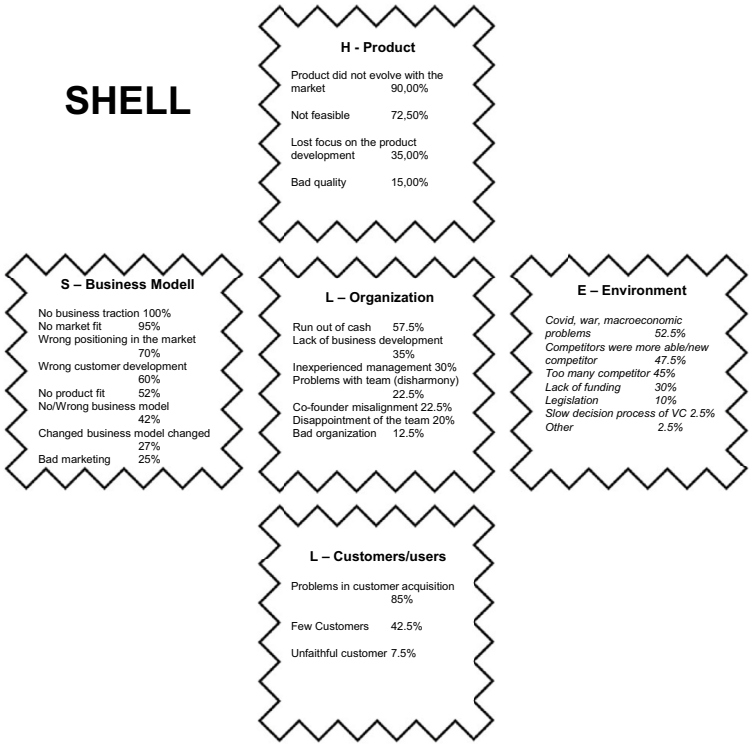



**Fig. 2** Extended version of the SHELL model and percentage of startups, where the variable was valid. This is the same block list diagram as Fig. 1. The only difference is that it includes the percentage of startups where the problem variable was valid (Source: own research, 2024)

The responses given for each variable were then aggregated to determine the most important factors. Based on all these points, it was already clear that common problems for the startups studied were:

- S: identifying the right market demand,
- H: The product did not involve the market,
- E: the impact of the global pandemic, war, and macroeconomy,
- L: attracting customers,
- L: running out of funding.

As a next step in our research, we added additional variables (Table 4) to get a more complete picture of the startups under study:

- variables related to the foundations (year of foundation, number of funders, etc.);
- achieved stage of their development (Minimum Viable Product (MVP), product-to-market fit, channel-to-market fit, growth/business traction);
- financial data (capital invested, cumulated revenue, market value, etc.).

The startups surveyed were founded between 2014 and 2021, and the average number of founders was 2.8 (only two startups were founded by 1 person, and the highest

number of founders was 7). The share of women among the founders was 35% of the businesses analyzed. On the question of whether the startup had participated in an accelerator program or had received support from an angel investor, the answer was yes in only 27.5% of cases. The number of investment rounds ranged from 1 to 4, with an average of 1.8 investments per company. Given the fact that the startups in the database were unsuccessful, it is not surprising that only 15% received some exit offer.

For the startups examined, the average time between founding and termination of activity was 4.25 years (between 1 and 9 years of operation). The average time from founding to MVP creation was 1.1 years, but only 35% of the database studied had reached the "product-to-market fit" stage. Only two startups reached the "channel-to-market fit" stage, and none of them achieved the "Growth/business traction" phase.

The average amount of capital invested was €397,000 (with a minimum of €28,000 and a maximum of €1,425,000). Almost three-quarters of the startups surveyed had some revenue, with an average cumulative revenue of €144,000. At voluntary liquidation, a positive market value could only be determined for two companies. 17.5% of the database surveyed had planned an international market presence from the start (and not just at a later stage).

#### Typology of failure factors and failed startups

To group failure factors, the following steps were used to reduce the initial large number of variables to 9 dichotomous and 3 metric variables:

1. Low variance variables were excluded from the 40 variables (where more than three-quarters of responses are identical).
2. Subsequently, narrowing was done based on covariance.
3. We used the principal component analysis in an attempt to maximize the KMO value.

Shen et al. (2023) explore the statistical significance of clustering, with a particular focus on high-dimensional, low sample size data. The study proposes new methods for assessing the reliability of clustering results, which may be useful for cluster analysis of startups. Volkovich et al. (2012) address the problem of cluster number selection and

**Table 4** Additional variables identified. Source: own research, 2024

| | Average | Minimum value | Maximum value |
|---|---|---|---|
| Number of founders | 2.8 | 1 | 7 |
| How many rounds of investment have you received from VCs? | 1.8 | 1 | 4 |
| Years between the date of foundation and termination | 4.275 | 1 | 9 |
| Years between the foundation and MVP date | 1.1 | 0 | 4 |
| Invested capital (tEUR) | 397.6 | 28 | 1425 |
| Cumulative revenue since foundation (tEUR) | 144 | 0 | 2765 |
| Are any of the founders female? | 35% | | |
| Were you participating in the startup accelerator program? Or have you had an angel investor? | 27.5% | | |
| Received any exit offers? | 15% | | |
| Was international market expansion a goal from the start? | 17.5% | | |

**Table 5** Shortlisted variables by the SHELL model and resource-based view/dynamic capability theory Source: own research, 2024

| Name of variable | Place in the SHELL model | Resource-based view/ dynamic capability theory |
|---|---|---|
| Not feasible | Product—H | Dynamic capability theory |
| No/wrong business model | Business model—S | Dynamic capability theory |
| Wrong positioning in the market | Business model—S | Dynamic capability theory |
| Bad promotion/advertisement | Business model—S | Dynamic capability theory |
| Lack of funding | Environment—E | Resource-based view |
| Few customers | Customer—L | Dynamic capability theory |
| Inexperienced management | Organization—L | Resource-based view |
| Problems with the team (disharmony) | Organization—L | Resource-based view |
| Lack of business development | Organization—L | Dynamic capability theory |
| Normalized invested capital | – | *Resource-based view* |
| How many rounds of investment have you received from VCs? | – | *Resource-based view* |
| Normalized cumulated turnover | – | No |

**Table 6** Examining the conditions for principal component analysis Source: own research based on SPSS output, 2024

| | |
|---|---|
| KMO and Bartlett's Test | |
| Kaiser–Meyer–Olkin measure of sampling adequacy | **0.622** |
| Bartlett's test of sphericity | |
| Approx. chi-square | 172,227 |
| D*f* | 66 |
| Sig | **0.0** |

Bold indicates Bartlett test is significant at 1% level

validation and highlight the importance of stability. Using the k-neighbor model, we measured the clustering quality and the correct determination of the cluster number. This is particularly important for small samples to be stable. The resulting variables are summarized in Table 5. The variables examined included financial and investment variables (shown in gray) and dichotomous variables describing the manifestation of a state or characteristic. All elements of the SHELL model are covered by at least one variable. Most of them reflect a dynamic capability rather than a company resource.

The KMO score is 0.623, indicating that the variables under study are more than moderately suitable for principal component analysis. The significance level of the Bartlett test confirmed that the principal component analysis was successfully performed (Table 6).

Based on Fig. 3, it can be seen that the eigenvalue falls below 1 for principal component 5. Hence, the creation of 4 principal components was considered appropriate. The variance explained was 70%; thus, the data loss was acceptable based on our analysis.

Table 7 presents the characteristics of the principal components.

In the first principal component, the variables with a high positive weight are mainly the amount of investment and the cumulative income from operations, as well as the number of investment rounds. Hence, this principal component is called Financial Performance. In the second principal component, lack of funding, lack of business development, and few customers are the main influencing factors. This is followed by

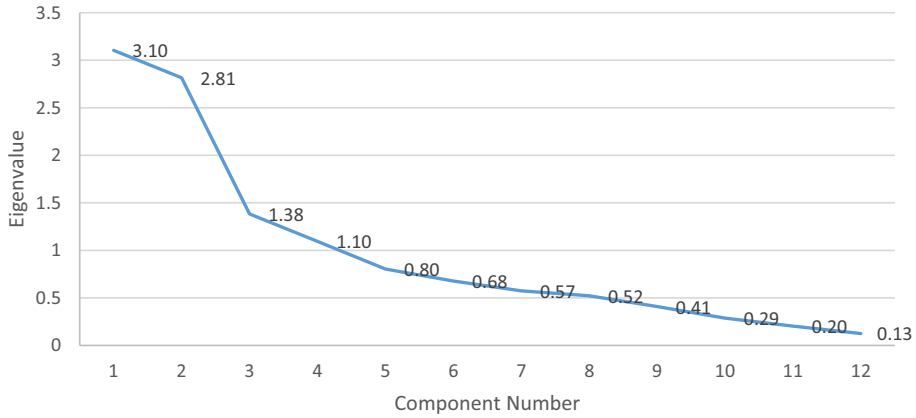



**Fig. 3** Visualization of the eigenvalues of principal component analysis with an elbow diagram. This is a line plot with the y-axis as the eigenvalue and the x-axis as the number of PCA components (Source: own research based on SPSS output, 2024)

**Table 7** Rotated component matrix. Source: own research based on SPSS output, 2024

| Rotated component matrix | PCA component | | | |
|---|---|---|---|---|
| **Variables** | **Financial performance** | **Market entry and customer acquisition** | **Inadequate business model** | **Management and product deficiency** |
| Normalized invested capital | ***0.889*** | 0.068 | −0.088 | −0.071 |
| How many rounds of investment have you received from VCs? | ***0.735*** | −0.166 | 0.266 | 0.228 |
| Normalized cumulated turnover | ***0.713*** | 0.178 | −0.099 | −0.034 |
| Lack of funding (environment—E) | 0.012 | **0.82** | 0.226 | 0.324 |
| Lack of business development (organization—L) | −0.11 | ***0.766*** | 0.392 | 0.153 |
| Few customers (customer—L) | 0.241 | ***0.734*** | −0.127 | −0.033 |
| Bad promotion/advertisement (business model—S) | −0.435 | ***0.537*** | 0.278 | 0.352 |
| Wrong positioning in the market (business model—S) | 0.152 | 0.082 | ***0.884*** | 0.06 |
| No/wrong business model (business model—S) | −0.195 | 0.129 | ***0.749*** | −0.011 |
| Problems with the team (organization—L) | 0.256 | 0.243 | −0.132 | ***0.824*** |
| Inexperienced management (organization—L) | −0.394 | 0.052 | 0.211 | ***0.552*** |
| Not feasible (product—H) | −0.514 | −0.532 | 0.161 | ***0.345*** |

Extraction method: principal component analysis

Rotation method: Varimax with Kaiser normalization

a Rotation converged in 11 iterations

bad promotion/advertisement. Therefore, this set of variables has been named Market Entry and Customer Acquisition. The third principal component is characterized by inadequate market positioning and a poor or inappropriate business model. Thus, this has been summarized as Inadequate Business Model. The fourth principal component included two organizational characteristics, i.e., disharmony within the team and inexperienced management, compounded by fundamental product problems.

Therefore, this variable group was named Management and Product Deficiency (Table 7).

To typify failed startups, we chose a ward-based hierarchical cluster analysis. The clusters created were selected by dendrogram. Our analysis produced three clusters that were well-separated from each other by analysis of variance based on the principal components. Our goal with cluster analysis is to identify the problem, so we look at the clustering from this perspective. We also evaluated the clusters using the original variables. In the results obtained, the values shown represent the proportion of the phenomenon from which the most important ones have been highlighted in gray (Table 8). There is a statistically detectable relationship between the variables (cluster and the respective explanatory variable), except for the variable inexperienced management. The visualizations of cluster differences (scatter plots or box plots) can be found in "Appendix".

The first cluster included 10 companies. The average scores show that Market Entry and Customer Acquisition were the main reasons for failure. Inadequate Business Model was also important to a lesser extent. To a minor level, Management and Product Deficiency also contributed to the failure. Negative Financial Performance indicates that they received relatively little funding and achieved virtually zero sales. An examination of the variables behind the main components confirms the former. Inadequate market positioning and weaknesses in business development were common to all firms in this cluster. Low customer numbers and problems with the business model are also common. The cluster was labeled "basic business model and financing problems." In this cluster, the founders could not manage the project adequately in several aspects besides the product development problems. The lack of dynamic capabilities was pronounced the most clearly in this group.

The second cluster has the largest number of elements (26 firms), where we see a lack of principal components. This suggests that the principal components and the variables remaining do not sufficiently explain the reasons for failure. Based on the variables examined (Table 9), infeasibility of the product or service is the only general characteristic of this group. Thus, the product idea failed. In addition, the related market positioning failure is also common. Hence, the term "failed product idea" has been applied to this group. We can consider this failure as a natural consequence of startup activity.

**Table 8** Results of the cluster analysis. Source: own research based on SPSS output, 2024

| Ward method | | PC1 financial performance | PC2 market entry and customer acquisition | PC3 inadequate business model | PC4 management and product deficiency |
|---|---|---|---|---|---|
| CL1 | N | 10 | 10 | 10 | 10 |
| | Mean | −0.33 | ***1**.26* | ***0.66*** | 0.27 |
| | Std. Deviation | 0.76 | 0.58 | 0.47 | 1.66 |
| CL2 | N | 26 | 26 | 26 | 26 |
| | Mean | −0.24 | −0.56 | −0.17 | −0.18 |
| | Std. Deviation | 0.50 | 0.60 | 1.06 | 0.59 |
| CL3 | N | 4 | 4 | 4 | 4 |
| | Mean | ***2.40*** | *0.**47*** | −0.55 | *0.**47*** |
| | Std. Deviation | 0.79 | 0.86 | 0.90 | 1.05 |

**Table 9** Results of the cluster analysis for the primary variables. Source: own research based on SPSS output, 2024

| | **Financial performance** | | | **Market Entry and Customer Acquisition** | | | | **Inadequate Business Model** | | **Management and Product Deficiency** | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Cluster 1 | | | | | | | | | | | | |
| Cluster 2 | | | | | | | | | | | | |
| Cluster 3 | | | | | | | | | | | | |
| | VC Investment round | Normalised invested capital | Normalized cumulated turnover | Lack of funding | Lack of business development | Few Customers | Bad promotion /advertisement | Wrong positioning in the market | No/Wrong Business Model | Inexperienced management | Problems with the team (disharmony) | Not feasible |
| Cluster 1 | 1,6 | 0,2 | 0,0 | **0,9** | **1,0** | **0,8** | **0,7** | **1,0** | **0,8** | 0,5 | 0,4 | 0,6 |
| Cluster 2 | 1,7 | 0,2 | 0,0 | 0,0 | 0,1 | 0,2 | 0,1 | 0,6 | 0,3 | 0,3 | 0,1 | **0,9** |
| Cluster 3 | **2,8** | **0,9** | **0,4** | 0,5 | 0,3 | **0,8** | - | **0,8** | - | - | **0,8** | - |

For the smallest, third cluster's companies (4) were the most successful in terms of investment and financial attraction. If we assess the problems, we can see problems in Market Entry and Customer Acquisition, as well as Management and Product Deficiency. Despite success in Financial Performance, faulty market positioning and few customers made this group unsuccessful. Therefore, we have labeled this group as "looked promising" due to its relative success.

The cluster analysis conducted in the study, including the dendrogram, ANOVA, and PROXSCAL analyses, revealed key factors influencing startup failures and highlighted the differences between the identified clusters.

The Ward hierarchical clustering analysis identified three clusters, which were visualized using a dendrogram. The dendrogram clearly illustrated the internal homogeneity of the clusters and their well-separated structures. The first cluster consisted of homogenous cases, the second showed moderate heterogeneity, while the third cluster was distinctly separated due to its significant inter-cluster distance. This visualization serves as a critical tool to display the hierarchical structure and relationships between cases.

The ANOVA analysis provided deeper insights into the differences between clusters along the principal components. The results showed significant differences in Financial Performance ($F = 44.35$, $p < 0.001$) and Market Entry and Customer Acquisition ($F = 33.95$, $p < 0.001$) between the clusters. However, no statistically significant differences were found for Inadequate Business Model ($F = 1.035$, $p = 0.365$) or Management and Product Deficiency ($F = 1.641$, $p = 0.208$). These findings suggest that inter-cluster differences are primarily driven by Financial Performance and Market Entry and Customer Acquisition, while business model and management problems play a less critical role in cluster separation.

The PROXSCAL analysis positioned the variables along two dimensions, facilitating a structured examination of the clusters. The first dimension highlighted differences between financial performance, and financial and customer problems, while the second dimension emphasized differences between Inadequate Business Model and Management and Product Deficiency. The fit indices—Normalized Raw Stress (0.1667) and D.A.F. (Dispersion Accounted For) (83.3%)—indicated moderately good results, showing that the model adequately explained the variance. The common space coordinates generated by PROXSCAL provided further insights into the relationships between variables and clusters.

The analyses demonstrated that the clusters are primarily distinguished by financial performance and customer acquisition issues, whereas business model and management problems have a lesser impact on differentiation. The interpretation of the results is strongly supported by the dendrogram, the ANOVA table, and the visual representation of the common space from PROXSCAL, all of which are essential figures to include in the article.

Cluster analysis can yield clusters with small differences due to several factors. First, the characteristics of the clustering algorithm can influence outcomes. For example, k-means may create overlapping clusters, particularly when a small number of clusters are predefined or the data is homogenous. This is due to the inherent dependency of K-means on preassigned cluster numbers and the homogeneity of the dataset (Hogan et al., 2009). Second, clusters may show small differences if many of the features used for clustering are irrelevant. In such cases, focusing on relevant features that are strongly associated with each cluster can enhance clustering results (Huang et al., 2021). Third, clustering stability can also impact the differences between clusters. Noisy data or small sample sizes may lead to less distinct clusters, making it harder to observe significant differences between them. This issue can be mitigated by applying normalization techniques (we did it) or using more robust clustering algorithms (Rebhahn et al., 2015). Lastly, small sample sizes with a large number of variables can make it difficult to form distinct clusters, especially when the data points are relatively homogeneous (Birley & Westhead, 1990). We excluded the relatively homogeneous variables.

#### The effect of life cycle on startup types and failure factors

Based on the literature, three main life cycle phases have been identified. The first is the minimum viable product (MVP) phase. The second is the product-to-market fit (PMF) phase, while the third phase is the channel-to-market (CHMF) phase. This is followed by the growth phase (business traction), but none of the samples reached this phase.

The data show that the highest proportion of firms in the MVP phase was in the third cluster. The highest failure rate was in the second cluster. The failure rate in the PMF phase was higher in the second and first clusters. It can be seen that half of the firms in the third cluster reached the CHMF phase. This finding also confirms the "looked promising" nature of the cluster.

The cross-table analysis found no significant relationship among clusters and variables describing life cycle success. However, the presentation of the results is relevant to our research (Table 10).

In the following, we run the cross-table analysis for the variables identified in the principal component analysis (Table 11). A significant relationship is indicated at the

**Table 10** The success of clusters by the life cycle. Source: own research based on SPSS output, 2024

| | Total *N* | MVP success (%) | PMF success (%) | CHMF success (%) |
|---|---|---|---|---|
| CL1 | 10 | 90 | 40 | 0 |
| CL2 | 26 | 80 | 30 | 0 |
| CL3 | 4 | 100 | 80 | 50 |

**Table 11** Failure as a variable in the different life cycles. Source: own research based on SPSS output, 2024

| | MVP failure | Sig | PMF failure | Sig | CHMF failure | Sig |
|---|---|---|---|---|---|---|
| Not feasible | ***1*** | | ***0.8*** | | ***0.8*** | ***0.02*** |
| No/Wrong Business Model | 0.6 | | 0.5 | | 0.4 | |
| Wrong positioning in the market | 0.8 | | 0.7 | | 0.7 | |
| Bad promotion/advertisement | 0.4 | | 0.3 | | 0.3 | |
| Lack of funding | 0.4 | | 0.2 | | 0.3 | ***0.03*** |
| Few Customers | 0 | ***0.04*** | 0.3 | ***0.04*** | 0.4 | |
| Inexperienced management | 0.6 | | 0.3 | | 0.3 | |
| Problems with the team (disharmony) | 0.2 | | 0.2 | | 0.2 | ***0.01*** |
| Lack of business development | 0.4 | | 0.3 | | 0.3 | |
| VC investment rounds | 1.6 | | 1.62 | | 1.74 | |
| Normalized invested capital | 0.04 | | 0.21 | | 0.25 | |
| Normalized cumulated turnover | 0.00 | | 0.01 | | 0.03 | |

Bold indicates the significant variables at 1% or 5 % level are highlighted in 'Sig' column

**Table 12** Failure as a variable in the different life cycles. Source: own research based on SPSS output, 2024

| Are there any women among the founders? | MVP_succ | | PMF_succ | | CHMF_succ | |
|---|---|---|---|---|---|---|
| | No | Yes | No | Yes | No | Yes |
| No | 4 | 22 | 20 | 6 | 26 | 0 |
| Yes | 1 | ***13*** | 6 | ***8*** | 12 | ***2*** |

value in the Sig column. The values in the table indicate the incidence rate for failure. The table identifies the causes of failure that are more prevalent in some life cycles and less prevalent in others. That is, as startups evolve, the probability of some problems occurring increases while others decrease. For example, the most prevalent variable of non-viable product failure occurred at 100% of MVP failures; with later stages, the prevalence of this variable decreased to 80%, where other problems also occurred.

- Decreasing importance with life cycle is not feasible variable (dynamic capability), no/wrong business model (dynamic capability), inexperienced management (dynamic capability);
- More significant with life cycle: few customers (dynamic capability), investment and financial indicators (resource).

The other variables could not be characterized by life cycle. Results show that dynamic capabilities are very important binding constraints in the early phases of startup growth.

#### The impact of gender on the survival of startups

In our study, we investigated whether female founders have an impact on the progression of a company. The cross-tables show that startups with female founders were more successful at all stages compared with cases with only male founders (Table 12).

**Table 13** Chi$^2$ test of gender and the PMF success variable. Source: own research based on SPSS output, 2024

| Gender vs PMF_succ | Value | df | Significance (2-sided) |
|---|---|---|---|
| Pearson Chi-Square | 4.642[a] | 1 | 0.031 |

[a] The test id significant at 5% level

**Table 14** F-test results for the exit offer and financial variables. Source: own research based on SPSS output, 2024

| | Sum of squares | df | Mean square | F | Sig |
|---|---|---|---|---|---|
| Invested capital | | | | | |
| Between groups | 500,357 | 1 | 500,357 | 3.754 | **0.05** |
| Within groups | 5,065,070 | 38 | 133,291 | | |
| Total | 5,565,428 | 39 | | | |
| Market value at termination | | | | | |
| Between groups | 2289 | 1 | 2289 | 6.701 | **0.014** |
| Within groups | 12,983 | 38 | 342 | | |
| Total | 15,272 | 39 | | | |
| How many rounds of investment have you received from VCs? | | | | | |
| Between groups | 3 | 1 | 3.459 | 6.939 | **0.012** |
| Within groups | 19 | 38 | 0.498 | | |
| Total | 22 | 39 | | | |

Bold indicates the significant variables at 1% or 5 % level are highlighted in 'Sig' column

In the MVP phase, the survival rate for startups with women is 93%; in the PMF phase, it is 53%, while in the CHMF phase, it is 14%. The same rates for male-only companies are 85%, 23%, and 0% respectively.

However, a link between the presence of a female founder and the life cycle variable was only statistically verified in the PMF phase, while no significant correlation was found between the other two phases (Table 13).

### Analysis of exit offers

If a business fails as a startup, i.e., it cannot grow as fast as investors and founders expect, it does not mean that the development it has created is completely worthless. This is illustrated by the fact that, before voluntary liquidation, startups that had already generated revenue and reached the PMF stage sometimes received a bid for their intellectual property (IP) that they had created (we call it an exit offer). In our analysis, we assessed whether the exit offer was influenced by the life cycle, clusters, and the presence of female founders. Among the factors examined, we found that the life cycle influenced the exit offer according to the chi$^2$ test. Thus, firms reaching the PMF[1] and CMF[2] stages received an exit offer. This is certainly not a surprising development. No significant difference was found with the other variables, but it was found that the second and third cluster members received an exit offer (4–2 ratio).

[1] $Chi^2 = 13.11$, $df = 1$, $p = 0.01$.

[2] $Chi^2 = 11.93$, $df = 1$, $p = 0.01$.

We also found that in 5 out of 6 exit offers, women were among the founders, and a significant relationship between the two variables was found ($chi^2 = 5.52$, $df = 1$, $p = 0.019$).

The companies that received an exit offer were also evaluated according to their financial characteristics. Our analysis revealed significant differences between companies receiving and not receiving an exit offer according to three aspects (Table 14).

As shown in the F-test, there is a difference in the capital invested, the market value at the exit, and the number of investment rounds between companies receiving and not receiving an exit offer (Table 15).

Since the exit variable is dichotomous, it is not possible to perform a post hoc test. Thus, we used a means-plot to evaluate the three variables. It can be seen that for the companies that received an offer, the average capital invested is double; these firms had a positive market value, and they received, on average, more rounds of investment.

## Discussion and conclusions

Failed startups are not a homogeneous group. The following types have been created:

1. The first group is characterized by "basic business model and financing problems", in which founders could not manage the business properly in several aspects, and in many cases, there were also fundamental problems with the product. The lack of dynamic capabilities is clearly identifiable in this group. In hindsight, from an investor perspective, this cluster should have been the least financed, and the founders should have prepared more before launching the project. The efficiency of the startup ecosystem can be improved by filtering out such projects, which bleed from multiple wounds. Incubation and/or mentoring programs could help these entrepreneurs to overcome these difficulties.
2. The second group had one general characteristic, the infeasibility of the product or service, which means that the product idea has failed. Thus, we call this type of startups as "failed product idea" group. Also common among them that there is an inadequate market positioning. These reasons for failure can be seen as natural in startup activity.
3. The third group of projects is in the category of "looked promising." These were more successful than average, raised more capital, generated revenues, and progressed further in the life cycle, but then their development was halted due to inadequate market positioning, few customers, and management problems.

**Table 15** The value of the averages in the different groups. Source: own research based on SPSS output, 2024

| Exit offer | No | Yes |
|---|---|---|
| Mean of invested capital (th EUR) | 349.7 | 663 |
| Mean of market value (th EUR) | 0.3 | 21.5 |
| Mean of No of investment rounds | 1.68 | 2.5 |

In our research, we identified a problem map that led to startup failures. Using principal component analysis, we shortlisted the problematic areas. The identified groups of these factors are:

1. Financial Performance (PCA1): its underlying variables are the amount of investment, the cumulative income from operations, and the number of investment rounds.
2. Market Entry and Customer Acquisition (PCA2) consist of lack of funding, lack of business development, few customers, and bad promotion/advertisement.
3. Inadequate Business Model (PCA3) includes wrong positioning in the market and no/wrong business model.
4. Management and Product Deficiency (PCA4) variables are inexperienced management, problems with the team (disharmony), and not feasible product.

PCA1 consists of traditional resource-based variables (and partly PCA2). PCA3 and 4 are pure groups of dynamic capabilities.

Cluster 1 is mostly characterized by PCA2 and 3. Cluster 2 there was a lack of principal components. However, two dynamic capability type variables defined its character. In the case of Cluster 3, resources were the most binding besides dynamic capabilities. Overall results support dynamic capability over RBV in a dynamic market. The SHELL model was an appropriate framework for detecting failure factors since all elements of the model were represented by at least one variable. In the study of Cantamessa et al. (2018), the lack of business development was the most pronounced predictor of startup failure, which was followed by no product/marketing mix, few customers, and no/wrong business model. These factors are all included in our results as well. All of them are dynamic capabilities.

Goswami et al. (2023) highlighted inadequate sales and marketing, and running out of money, which is our first group of causes. Product viability risks are also revealed in the literature (Bednár & Tarišková, 2017; Chakraborty et al., 2023) as well as lack of funding (Díaz-Santamaría & Bulchand-Gidumal, 2021).

As startups evolve, the probability of some problems occurring increases (few customers) while others decrease (not feasible, no/wrong business model, and inexperienced management). All of them are dynamic capabilities. Financial and investment indicators improve (resources) as the life cycle progresses. Results show that dynamic capabilities are very important binding constraints in the early phases of startup growth. This calls for further research into the measurement and development of dynamic capabilities in early-stage businesses.

Women-owned businesses are more likely to survive each phase. In the sample, there is statistically significant evidence that startups with female founders in the PMF phase are more successful in making it through this life cycle. This also induces startups to be more likely to receive an exit offer. The results confirm the branch of

the literature (Kee & Rahman, 2018; Ong et al., 2016) which found a positive relationship between startup performance and female entrepreneurship. In most countries in the world, there is an aging society, where the workforce is becoming more and more scarce. In such an environment, efficiently involving women in entrepreneurship is becoming an increasingly important economic topic, no longer just a social issue.

The existence of an exit offer is an indicator of viability, which is typically found in companies with more successful financial parameters. Our study revealed that the factors influencing the exit offer depend on the life cycle and the gender composition among founders. Other factors that influence exit offers are the level of funding and the market value. Nevertheless, they were not capable of the rapid growth expected from startups.

## Policy implications

Our analysis allows us to draw some policy implications. Startups and their investors face a high probability of failure. If the startup's product idea is ultimately not justified by market demand, we consider this to be a natural cause of failure. At the same time, a group of failed startups can be identified, where the founders failed to manage the business in several important aspects: they struggled with product, customer, management, and sales at the same time. On the one hand, we suggest specific policy interventions to help startups strengthen dynamic capabilities, such as innovation and corporate foresight training programs (Schwarz et al., 2020), mentorship access, or ecosystem support. On the other hand, strengthening project screening standards can also reduce the rate of such projects. In this way, the efficiency of the startup ecosystem might improve.

Our sample consisted exclusively of startups in the ICT sector. Despite this, the more gender-balanced management proved to be relatively more successful. This suggests that both founders and investors should encourage the involvement of female founders in projects despite the conventions and the numerous obstacles listed in the literature. We support the view of OECD (2021) to further develop policies to address gender disparities in entrepreneurship, such as overarching policy frameworks for women's entrepreneurship, tailored funding programs, mentorship initiatives for female founders, and work-life balance support mechanisms. Increasing participation of women in startups might increase the efficiency and the success rate of startup ecosystems.

## Appendix: Visualizations of cluster differences

See Figs. 4, 5, Table 16.

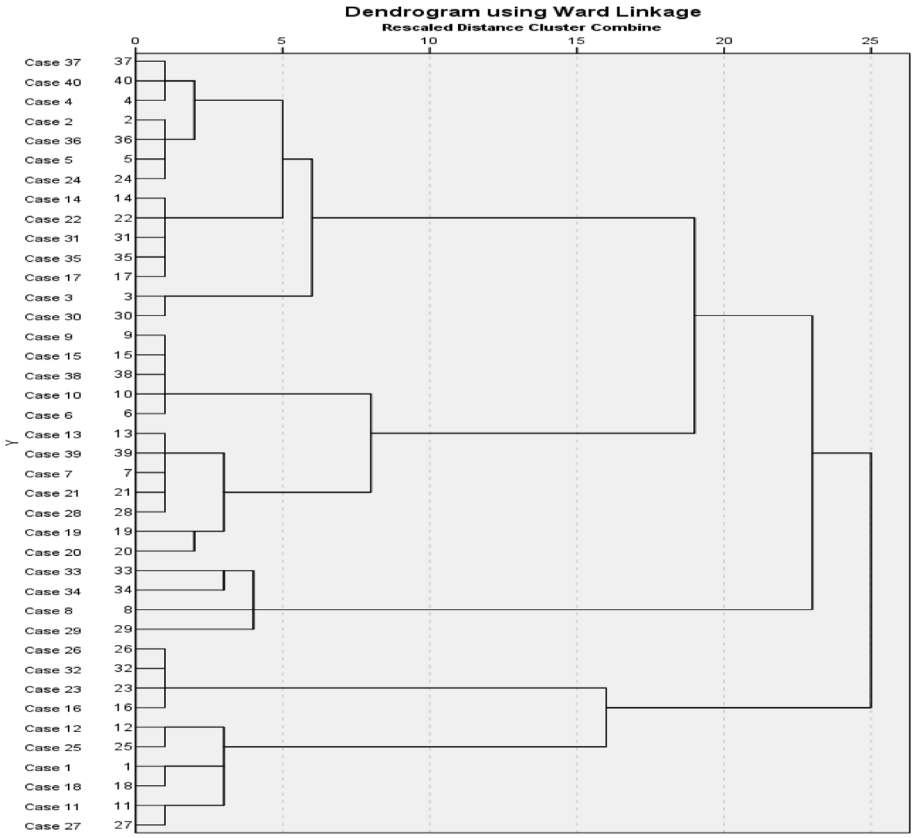



**Fig. 4** Cluster analysis dendrogram

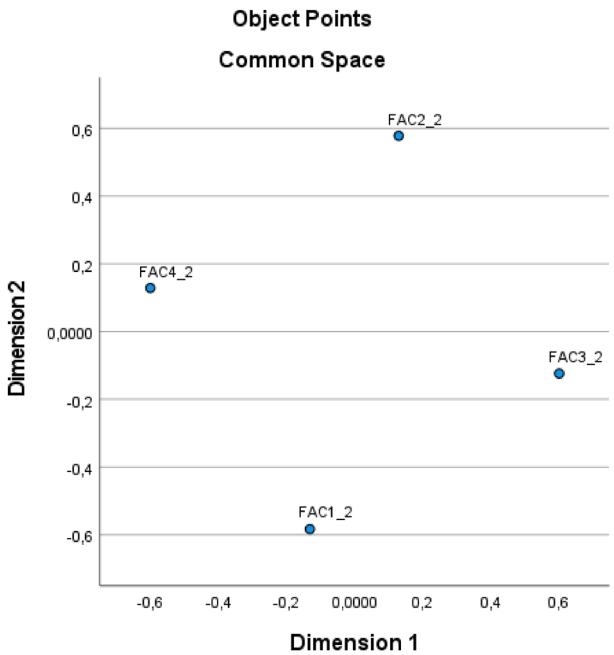



**Fig. 5** Proximities in the variables

**Table 16** ANOVA of the clusters and principal components

| Principal component | Sum of squares | *df* | Mean square | *F* | Sig |
|---|---|---|---|---|---|
| Financial performance | | | | | |
| Between groups | 27.030 | 2 | 13.515 | 44.350 | 0.000 |
| Within groups | 10.970 | 36 | 0.305 | | |
| Total | 38.000 | 38 | | | |
| Market entry and customer acquisition | | | | | |
| Between groups | 24.834 | 2 | 12.417 | 33.950 | 0.000 |
| Within groups | 13.166 | 36 | 0.366 | | |
| Total | 38.000 | 38 | | | |
| Inadequate business model | | | | | |
| Between groups | 2.067 | 2 | 1.033 | 1.035 | 0.365 |
| Within groups | 35.933 | 36 | 0.998 | | |
| Total | 38.000 | 38 | | | |
| Management and product deficiency | | | | | |
| Between groups | 3.175 | 2 | 1.587 | 1.641 | 0.208 |
| Within groups | 34.825 | 36 | 0.967 | | |
| Total | 38.000 | 38 | | | |

**Acknowledgements**
Not applicable.

**Author contributions**
András Bethlendi devised the project, the main conceptual ideas, and the proof outline. He supervised the work. He prepared Sects. Introduction, Discussion and conclusions, and Policy implications. Árpád Szőcs and Szilárd Hegedüs did the interviews and data collection. Árpád Szőcs summarized the literature (Sect. Literature review). Szilárd Hegedüs designed the model and the computational framework and analyzed the data (Sects. Data and methodology–Results).

**Funding**
Open access funding provided by Budapest University of Technology and Economics. András Bethlendi was supported by the Sustainable Development and Technologies National Programme of the Hungarian Academy of Sciences (FFT NP FTA). Szilárd Hegedűs is a scholarship recipient of the Sándor Demján Foundation.

**Availability of data and materials**
The datasets generated and/or analyzed during the current study are not publicly available due to their corporate-sensitive statements.

## Declarations

**Competing interests**
We have no conflict of interest to disclose.

Received: 23 October 2024 Accepted: 23 February 2025
Published online: 28 March 2025

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